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Problem 1

1. **Title: Conversion Calculator Inches to Centimetres:**
   Objective: The objective of this problem is to develop a simple Android application that converts a given value from Inches to Centimetres using user input and button click functionality.
   Problem Statement: Create an Android application with an input field where the user enters a value in Inches. Upon clicking the Convert button, the application should display the converted value in Centimetres.
   Algorithm:
   Step 1: Start
   Step 2: User enters value in inches
   Step 3: User clicks Convert button
   Step 4: Read the input value
   Step 5: Convert inches to centimetres (cm = inches × 2.54)
   Step 6: Display the result
   Step 7: Stop
   Flow Chart: Start → Input Inches → Click Convert → Calculate → Display Result → End
   Source Code:
   - activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">
    <EditText
        android:id="@+id/etInches"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter value in Inches"
        android:inputType="numberDecimal" />
    <Button
        android:id="@+id/btnConvert"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Convert"
        android:layout_marginTop="16dp" />
    <TextView
        android:id="@+id/tvResult"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Result will be shown here"
        android:textSize="18sp"
        android:layout_marginTop="16dp" />
```

```
</LinearLayout>
```

- MainActivity.java

```
package com.example.inchestocm;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
public class MainActivity extends AppCompatActivity {
    EditText etInches;
    Button btnConvert;
    TextView tvResult;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        etInches = findViewById(R.id.etInches);
        btnConvert = findViewById(R.id.btnConvert);
        tvResult = findViewById(R.id.tvResult);
        btnConvert.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String input = etInches.getText().toString();
                if (!input.isEmpty()) {
                    double inches = Double.parseDouble(input);
                    double centimetres = inches * 2.54;
                    tvResult.setText("Centimetres: " + centimetres);
                } else {
                    tvResult.setText("Please enter a value");
                }
            }
        });
    }
}
```

App Screenshots:

Figure 1: App Home Screen

Figure 2: Inches Input Screen

Figure 3: Converted Output Screen

Problem 2

2. **Title: Tap Counter Android Application**
   Objective: To develop a simple Android application that counts the number of times a button is tapped and displays the count on the screen.
   Problem Statement: Create an Android application with a button labelled “Tap me!” and a TextView to display the number of taps.
   Each time the button is tapped, the count should increase by one.
   Expected Output: The application should display the number of times the button has been tapped.
   Software & Tools Used:
   - Android Studio
   - Java Programming Language
   - XML for UI Design
   - Android Emulator / Physical Android Device

   Algorithms:
   Step 1: Start the application
   Step 2: Initialize tap count to zero
   Step 3: Display button labelled "Tap me!"
   Step 4: Display TextView showing tap count
   Step 5: When button is clicked:
   a. Increment tap count by 1
   b. Update TextView with new count
   Step 6: Repeat Step 5 for each tap
   Step 7: Stop
   Flow Chart: Start → Initialize Count=0 → Display Button & Text View → Button Clicked? → Yes → Increase Count by 1 → Display Updated Count → End

   Source Code:

   - activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">
    <Button
        android:id="@+id/btnTap"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Tap me!" />
    <TextView
        android:id="@+id/tvCount"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Taps: 0"
        android:textSize="18sp"
        android:layout_marginTop="20dp" />
</LinearLayout>
```

- MainActivity.java

```
package com.example.tapcounter;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;
public class MainActivity extends AppCompatActivity {
    Button btnTap;
    TextView tvCount;
    int count = 0;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        btnTap = findViewById(R.id.btnTap);
        tvCount = findViewById(R.id.tvCount);
        btnTap.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                count++;
                tvCount.setText("Taps: " + count);
            }
        });
    }
}
```

App Screenshots:

Figure 1: App Input Screen

Figure 2: App Output Screen

Problem 3

3. **Title: BMI Calculator based on Height & Weight**

   Objective: To create an Android application that calculates the Body Mass Index (BMI) of a user based on weight and height inputs, and determines the corresponding weight category (Underweight, Normal, Overweight, Obese).

   Problem Statement: Design and develop an Android app with the following requirements:

   - Two input fields for the user to enter weight (in kilograms) and height (in meters).
   - A button labelled "Calculate BMI".
   - When the button is pressed, the app should display:
     - The BMI value (rounded to 2 decimal places).
     - The BMI Category based on the BMI value:
       - Underweight (< 18.5)
       - Normal (18.5 – 24.9)
       - Overweight (25 – 29**.9)**
       - Obese (≥ 30)

   Source Code:

   - activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp"
    android:gravity="center_horizontal">
    <TextView
        android:id="@+id/titleText"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="BMI Calculator"
        android:textSize="24sp"
        android:textStyle="bold"
        android:layout_marginBottom="20dp"/>
    <EditText
        android:id="@+id/weightInput"
        android:layout_width="match_parent"
```

```
        android:layout_height="wrap_content"
        android:hint="Weight (kg)"
        android:inputType="numberDecimal"
        android:layout_marginBottom="15dp"/>
    <EditText
        android:id="@+id/heightInput"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Height (m)"
        android:inputType="numberDecimal"
        android:layout_marginBottom="20dp"/>
    <Button
        android:id="@+id/calculateBtn"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Calculate BMI"
        android:padding="10dp"
        android:layout_marginBottom="20dp"/>
    <TextView
        android:id="@+id/bmiResult"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="BMI: "
        android:textSize="18sp"
        android:layout_marginBottom="10dp"/>
    <TextView
        android:id="@+id/categoryResult"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Category: "
        android:textSize="18sp"/>
</LinearLayout>
```

- MainActivity.kt

```
package com.example.bmicalculator
import android.os.Bundle
import android.widget.Button
import android.widget.EditText
```

```
import android.widget.TextView
import android.widget.Toast
import androidx.appcompat.app.AppCompatActivity
class MainActivity : AppCompatActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)
        val weightInput = findViewById<EditText>(R.id.weightInput)
        val heightInput = findViewById<EditText>(R.id.heightInput)
        val calculateBtn = findViewById<Button>(R.id.calculateBtn)
        val bmiResult = findViewById<TextView>(R.id.bmiResult)
        val categoryResult = findViewById<TextView>(R.id.categoryResult)
        calculateBtn.setOnClickListener {
            val weightText = weightInput.text.toString()
            val heightText = heightInput.text.toString()
            if (weightText.isEmpty() || heightText.isEmpty()) {
                Toast.makeText(this, "Please enter weight and height",
Toast.LENGTH_SHORT).show()
                return@setOnClickListener
            }
            val weight = weightText.toFloat()
            val height = heightText.toFloat()
            if (height <= 0 || weight <= 0) {
                Toast.makeText(this, "Enter valid positive numbers",
Toast.LENGTH_SHORT).show()
                return@setOnClickListener
            }
            val bmi = weight / (height * height)
            bmiResult.text = "BMI: %.2f".format(bmi)
            categoryResult.text = "Category: ${getBMICategory(bmi)}"
        }
    }
    private fun getBMICategory(bmi: Float): String {
        return when {
            bmi < 18.5 -> "Underweight"
            bmi < 25 -> "Normal"
            bmi < 30 -> "Overweight"
```

```
            else -> "Obese"
        }
    }
}
```

Explanation:

- The user enters weight and height values.
- On pressing **the Calculate BMI button:**
  - BMI is calculated using the formula

    BMI = weight / height^2
  - The app determines on the screen under BMI Result and Category Result

App Layout / Screenshots:

- **Home Screen:** Input fields for weight and height. Calculate BMI button
- **After Entering Data:** Weight = 70kg , Height = 1.75m
- **After Pressing Calculate BMI:** BMI results: 22.86, Category: Normal

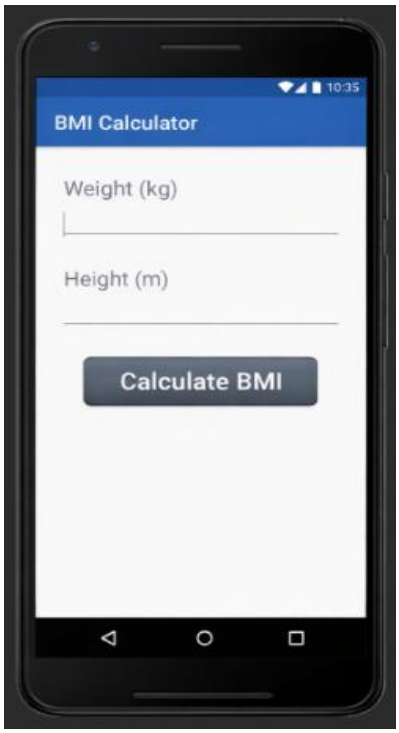

Figure 1: Home Screen

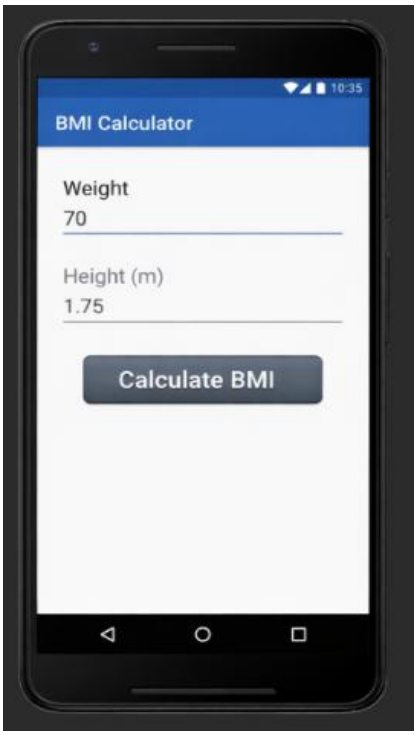

Figure 2: After Entering Data

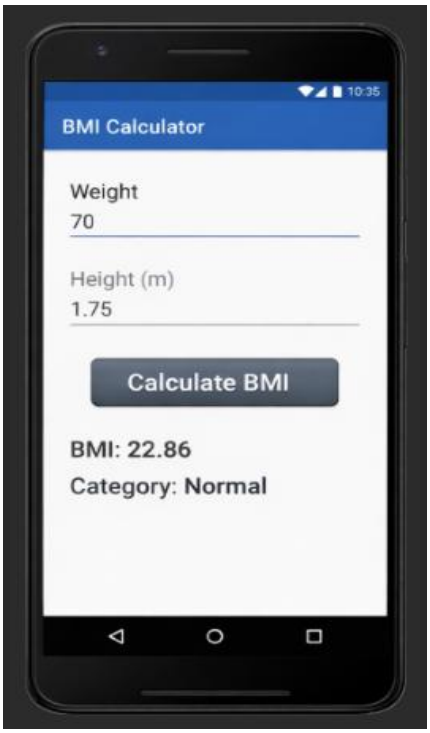

Figure 3: After Pressing BMI

Problem 4

4. **Title: Navigate to Other Activities using Explicit Intents**

   Objective: To create a simple Android application demonstrating navigation between activities using Explicit Intents.

   Problem Statement: Design and implement an Android app with the following requirements:

   - The app contains two activities MainActivity and SecondActivity.
   - Each Activity contains a button
     - MainActivty button opens SecondActivity
     - SecondActivity button opens MainActivity
   - The app demonstrates navigation between activities using explicit intents

   Explanation Working:

   - The app contains two activities: MainActivity and SecondActivity.
   - When the user clicks the button in MainActivity, an explicit intent starts SecondActivity.
   - Similarly, clicking the button in SecondActivity starts MainActivity using an explicit intent.
   - This demonstrates activity navigation and explicit intents in Android.

   Source Code:

   - **MainActivity:** MainActivity.kt

```
package com.example.intentdemo
import android.content.Intent
import android.os.Bundle
import android.widget.Button
import androidx.appcompat.app.AppCompatActivity
class MainActivity : AppCompatActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)
        val buttonGoSecond = findViewById<Button>(R.id.buttonGoSecond)
        buttonGoSecond.setOnClickListener {
            // Explicit Intent to start SecondActivity
            val intent = Intent(this, SecondActivity::class.java)
            startActivity(intent)
        }
    }
}
```

- **SecondActivity:** SecondActivity.kt

```
package com.example.intentdemo
import android.content.Intent
import android.os.Bundle
import android.widget.Button
import androidx.appcompat.app.AppCompatActivity
class SecondActivity : AppCompatActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_second)
        val buttonGoMain = findViewById<Button>(R.id.buttonGoMain)
        buttonGoMain.setOnClickListener {
            // Explicit Intent to go back to MainActivity
            val intent = Intent(this, MainActivity::class.java)
            startActivity(intent)
        }
    }
}
```

- **Layout Files:** activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    android:padding="20dp">
    <Button
        android:id="@+id/buttonGoSecond"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Go to Second Activity"
        android:padding="10dp"/>
</LinearLayout>
```

- **Layout Files:** activity_second.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    android:padding="20dp">
    <Button
        android:id="@+id/buttonGoMain"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Go to Main Activity"
        android:padding="10dp"/>
</LinearLayout>
```

- **Manifest Updates:** AndroidManifest.xml

```
<application
    android:allowBackup="true"
    android:icon="@mipmap/ic_launcher"
    android:label="@string/app_name"
    android:roundIcon="@mipmap/ic_launcher_round"
    android:supportsRtl="true"
    android:theme="@style/Theme.IntentDemo">
    <activity android:name=".SecondActivity" />
    <activity android:name=".MainActivity">
        <intent-filter>
            <action android:name="android.intent.action.MAIN" />
            <category android:name="android.intent.category.LAUNCHER" />
        </intent-filter>
    </activity>
</application>
```

Expected Output / Screenshot:

- Launch App MainActivity appears with a button
- Click “Go to Second Activity” SecondActivity opens
- Click “Go to Main Activity” MainActivity opens

Figure 1: Home Screen

Figure 2: Second Activity

Figure 3: Main Activity

Problem 5

5. **Title: Calculator which performs Arithmetic Operations**

Objective: To develop a simple Android application that performs basic arithmetic operations: addition, subtraction, multiplication, and division.

Problem Statement: Design and implement an Android app with the following requirements:

- Two input fields for the user to enter numbers.
- Buttons for the four arithmetic operations: Addition (+), Subtraction (-), Multiplication (×), Division (÷).
- A "Calculate" button to perform the selected operation.
- Display the result of the selected operation below the inputs.

Explanation Working:

- User enters two numbers in the EditText fields.
- Selects an arithmetic operation via RadioButton.
- Clicks the Calculate button.
- App calculates the result and displays it in the TextView
- Handles division by zero gracefully with a Toast message.

Source Code:

- activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
```

```
xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:orientation="vertical"
android:padding="20dp"
android:gravity="center_horizontal">
<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Simple Calculator"
    android:textSize="24sp"
    android:textStyle="bold"
    android:layout_marginBottom="20dp"/>
<EditText
    android:id="@+id/number1"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter first number"
    android:inputType="numberDecimal"
    android:layout_marginBottom="10dp"/>
<EditText
    android:id="@+id/number2"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter second number"
    android:inputType="numberDecimal"
    android:layout_marginBottom="20dp"/>
<RadioGroup
    android:id="@+id/operationGroup"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:layout_marginBottom="20dp">
    <RadioButton
        android:id="@+id/addition"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
```

```
            android:text="+" />

        <RadioButton
            android:id="@+id/subtraction"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="-" />
        <RadioButton
            android:id="@+id/multiplication"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="×" />
        <RadioButton
            android:id="@+id/division"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="÷" />
    </RadioGroup>
    <Button
        android:id="@+id/calculateBtn"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Calculate"
        android:padding="10dp"
        android:layout_marginBottom="20dp"/>
    <TextView
        android:id="@+id/resultView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Result: "
        android:textSize="18sp"/>
</LinearLayout>
```

- MainActivity.kt

```
package com.example.simplecalculator
import android.os.Bundle
import android.widget.*
import androidx.appcompat.app.AppCompatActivity
```

```
class MainActivity : AppCompatActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)
        val number1 = findViewById<EditText>(R.id.number1)
        val number2 = findViewById<EditText>(R.id.number2)
        val operationGroup = findViewById<RadioGroup>(R.id.operationGroup)
        val calculateBtn = findViewById<Button>(R.id.calculateBtn)
        val resultView = findViewById<TextView>(R.id.resultView)
        calculateBtn.setOnClickListener {
            val num1Text = number1.text.toString()
            val num2Text = number2.text.toString()
            if (num1Text.isEmpty() || num2Text.isEmpty()) {
                Toast.makeText(this, "Please enter both numbers",
Toast.LENGTH_SHORT).show()
                return@setOnClickListener
            }
            val num1 = num1Text.toDouble()
            val num2 = num2Text.toDouble()
            val selectedOperation = operationGroup.checkedRadioButtonId
            val result = when(selectedOperation) {
                R.id.addition -> num1 + num2
                R.id.subtraction -> num1 - num2
                R.id.multiplication -> num1 * num2
                R.id.division -> {
                    if (num2 == 0.0) {
                        Toast.makeText(this, "Cannot divide by zero",
Toast.LENGTH_SHORT).show()
                        return@setOnClickListener
                    } else num1 / num2
                }
                else -> {
                    Toast.makeText(this, "Please select an operation",
Toast.LENGTH_SHORT).show()
                    return@setOnClickListener
                }
            }
```

```
            val operationSymbol = when(selectedOperation) {
                R.id.addition -> "+"
                R.id.subtraction -> "-"
                R.id.multiplication -> "×"
                R.id.division -> "÷"
                else -> ""
            }
            resultView.text = "Result: $num1 $operationSymbol $num2 = $result"
        }
    }
}
```

Expected Output / Screenshots:

Input 1	Input 2	Operation	Output
10	5	+	10 + 5 = 15
10	5	-	10 – 5 = 5
10	5	x	10 × 5 = 50
10	5	÷	10 ÷ 5 = 2

Figure 1: Home Screen

Figure 2: After Entering Values

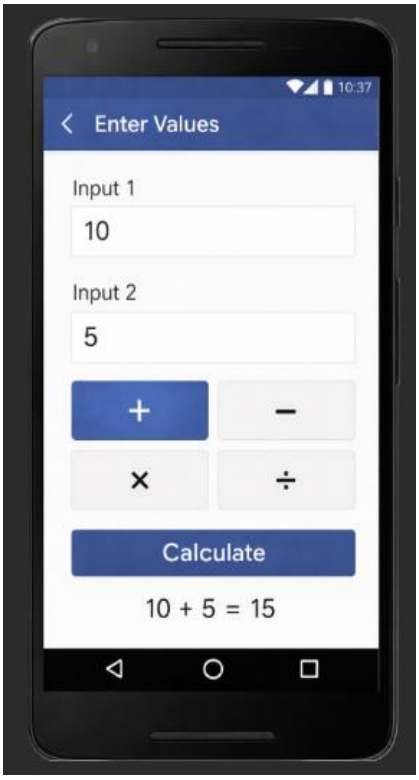

Figure 3: Using Addition

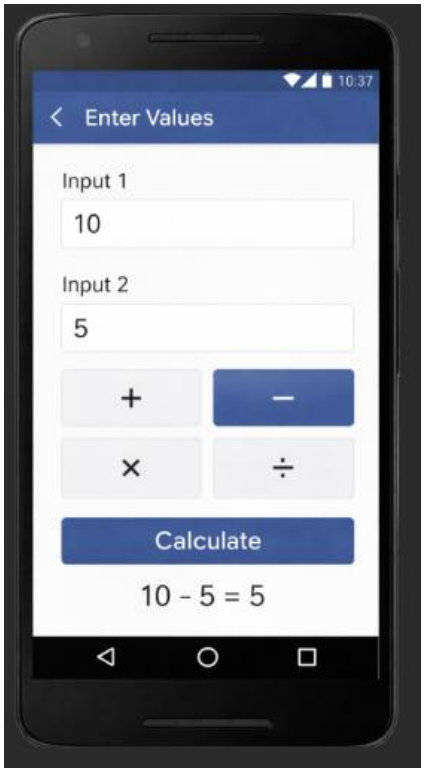

Figure 4: Using Subtraction

Figure 5: Using Multiplication

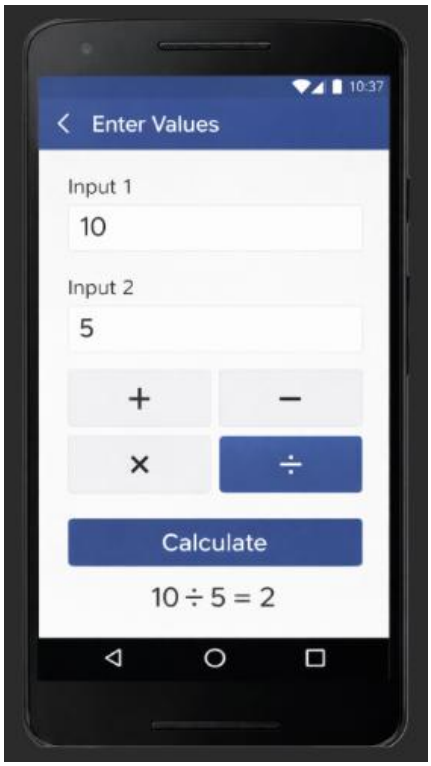

Figure 6: Using Division